Chapter 1: Numerical Methods

We introduce a locally mass conserving numerical framework. We solve the static mechanics system with conforming mixed finite element method, which is adjusted to be locally mass conservative. We solve the transport system with finite volume scheme, which is a mass conservative scheme.

1.1 Quadrilateral Mesh

We consider a bounded computational domain $\Omega \subset \mathbb{R}^2$. Here, $\partial\Omega$ denotes the Lipschitz-continuous boundary of the domain. Let $\mathcal{T}_h$ be a conforming mesh of quadrilaterals covering Ω. The quadrilateral $E \in \mathcal{T}_h$ is referred as element in finite element context, while referred as cell in finite volume context. Each quadrilateral E is assumed to be strictly convex, not degenerate into a triangle, line or point. For ease of implementation, we use a logically rectangular mesh, where each element(cell) is indexed using Cartesian coordinates $\{i, j\}$. We show a reference element(cell) $\hat{E}$ and a general quadrilateral element(cell) Q in Figure 1.1. There exists a non-affine bijective and smooth map $\mathbf{F}_{\hat{E}} : \hat{E} \to E$ mapping the vertices of $\hat{E}$ to the vertices of E.

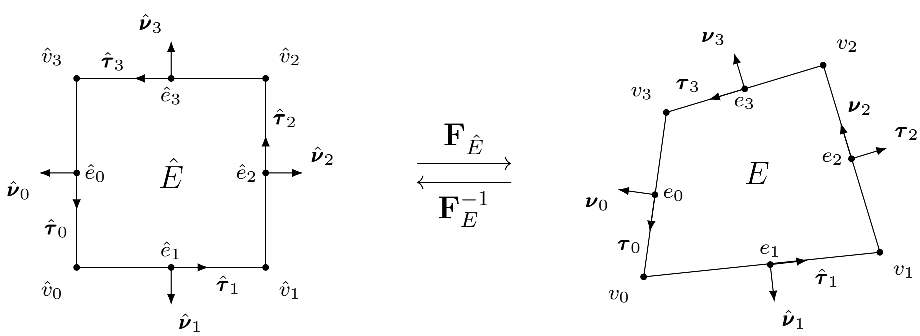

Figure 1.1: An exhibition of a reference square and a general non-degenerate quadrilateral element(cell).

The edges of element E are denoted by e_i, $i \in \{0,1,2,3\}$ in the counterclockwise direction. The vertices are represented as $v_i = e_i \cap e_{i+1}$ in the sense of modulo. Here, $\boldsymbol{\nu}_i$ and $\boldsymbol{\tau}_i$ are used to represent the unit normal and the unit tangent vector respectively. The unit normal vector $\boldsymbol{\nu}_i$ points outerwards the quadrilateral. The tangent $\boldsymbol{\tau}_i$ is generated by rotating $\boldsymbol{\nu}_i$ counterclockwise by right-hand rule.

In Figure 1.2, we illustrate a full logically rectangular qusi-uniform quadrilateral mesh $\mathcal{T}_h$, which is generated by randomly distort the interior vertex of a square mesh.

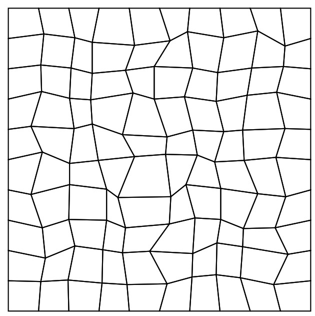

Figure 1.2: An exhibition of a full logically rectangular qusi-uniform quadrilateral mesh $\mathcal{Q}_h$.

1.1.1 Shape Regularity

We give an official definition of uniform shape regular of a quadrilateral mesh. For any quadrilateral $E \in \mathcal{T}_h$, we can get a set of four triangles T_i, $\quad i \in \{0,1,2,3\}$. We form the triangle T_i by excluding vertex v_i from the quadrilaterl E, and use the remaining three vertex. We define the following parameters as

$$\begin{aligned} h_E &= \text{diameter of} E, \\ \rho_E &= 2 \min_{1 \le i \le 4} \{\text{diameter of largest circle inscribed in } T_i\}. \end{aligned} \tag{1.1}$$

We consider a mesh partion $\mathcal{T}_h$ is uniformly shape regular if there exists a shape regularity parameter σ_* independent of $\mathcal{T}_h$, such that the ratio

$$\rho_E / h_E \ge \sigma_* > 0, \tag{1.2}$$

for all quadrilaterals $E \in \mathcal{T}_h$.

1.2 Direct Serendipity Space

Serendipity element have the same accuracy of approximation with classical tensor product element while use smaller number of degrees of freedom. However, the classical serendipity space suffer from poor approximation on quadrilaterals. We define local shape functions directly on quadrilaterals according to Arbogast et al. (2022). H^1 and $H(\text{div})$ conforming direct mixed finite elemenent space are constructed with the direct serendipity function space.

To begin with, we introduce some notations and distance auxilliary functions. Let $\mathbb{P}_r(\omega)$ denote the space of polynomials of degree up to r on $\omega \subset \mathbb{R}^d$. The dimension of $\mathbb{P}_r$ can be calculated as

$$\dim\mathbb{P}_r(\mathbb{R}^d) = \binom{r+2}{2} = \frac{(r+d)!}{r!d!}. \tag{1.3}$$

We define the linear polynomial $\lambda_i(\mathbf{x})$ giving the distance of $\mathbf{x} \in \mathbb{R}^2$ to edge e_i as

$$\lambda_i(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_i^*) \cdot \boldsymbol{\nu}_i, \quad i = 0, 1, 2, 3, \tag{1.4}$$

where $\mathbf{x}_i^* \in e_i$ is any point on the edge e_i. The value of distance function λ_i keeps positive as long as $\boldsymbol{x}$ is in the interior of the quadrilateral. We define two additional distance functions giving the distance of $\mathbf{x} \in \mathbb{R}^2$ to diagonal line d_i joining v_i with v_{i+2}, $i \in 0, 1$. Let $\nu_{d,i}$ be the unit normal vector to the diagonal line d_i. The direction of the $\nu_{d,i}$ is defined by counterclockwise rotating of the vector pointing from v_i to v_{i+2}. The resulting diagonal distance function is defined as

$$\lambda_{d,i}(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_{d,i}^*) \cdot \boldsymbol{\nu}_{d,i}, \quad i = 1, 2, \tag{1.5}$$

where $\mathbf{x}_{d,i}^*$ is any point on the diagonal line d_i. For simplicity, we use corner v_i in later computation.

Now we present set of nodal basis functions in first and second order direct serendipity space $\mathcal{DS}_1$ and $\mathcal{DS}_2$ defined by Arbogast et al. (2022).

1.2.1 Direct Serendipity Space $\mathcal{DS}_2$

We supplement polynomial space $\mathbb{P}_r(E), r \geq 2$ with two linearly independent functions, $\phi_{s,1}(\mathbf{x})$ and $\phi_{s,2}(\mathbf{x})$. We denote the supplemental function space as $\mathbb{S}_r^{\mathcal{DS}}(E) = \text{span}\{\phi_{s,1}, \phi_{s,2}\}, \quad r \geq 2$. We present the second order direct serendipity space as

$$\mathcal{DS}_2(E) = \mathbb{P}_2(E) \oplus \mathbb{S}_2^{\mathcal{DS}}(E). \tag{1.6}$$

The supplemental functions are defined as

$$\phi_{s,i}(\mathbf{x}) = \lambda_{i+1}(\mathbf{x})\lambda_{i+2}(\mathbf{x})R_{i,i+2}(\mathbf{x}), \quad i \in \{1, 2\}, \tag{1.7}$$

where i is in the sense of modulo. We use a simple definition for rational function $R_{i,i+2}$ as

$$R_{i,i+2}(\mathbf{x}) = \frac{\lambda_i(\mathbf{x}) - \lambda_{i+2}(\mathbf{x})}{\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x})}, \tag{1.8}$$

which satisfies the conditions as

$$R_{i,i+2}|_{e_i} = -1, \quad R_{i,i+2}|_{e_{i+2}} = 1. \tag{1.9}$$

We continue to define

$$R_i(\mathbf{x}) = \frac{1}{2}\Big(1 - (-1)^{\lfloor i/2 \rfloor} R_{i,i+2}(\mathbf{x})\Big) = \frac{\lambda_{i+2}(\mathbf{x})}{\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x})}, \tag{1.10}$$

where i is understood $\pmod 2$. Here, $R_i|_{e_i} = 1$, and $R_i|_{e_{i+2}} = 0$ by definition. In Figure 1.3, we draw R_0 on a general quadrilateral, which evaluates 1 on the edge e_0, and evaluates 0 on the opposite edge e_2.

The edge nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi_{e,i}(\mathbf{x}) = \frac{\lambda_{i+1}(\mathbf{x})\lambda_{i+3}(\mathbf{x})R_i(\mathbf{x})}{\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{e,i})}, \tag{1.11}$$

where $\mathbf{x}_{e,i}$ is the middle point of the edge e_i, the i index is understood in the sense of modulo of 4. In Figure 1.4, we present $\varphi_{e,0}$ on a general quadrilateral.

The vertex nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi_{v,i}(\mathbf{x}) = \frac{\lambda_{i+2}(\mathbf{x})\lambda_{i+3}(\mathbf{x}) - \sum_{k=i}^{i+1} \lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi_{e,i}(\mathbf{x})}{\lambda_{i+2}(\mathbf{x}_{v,i})\lambda_{i+3}(\mathbf{x}_{v,i}) - \sum_{k=i}^{i+1} \lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi_{e,i}(\mathbf{x}_{v,i})}, \tag{1.12}$$

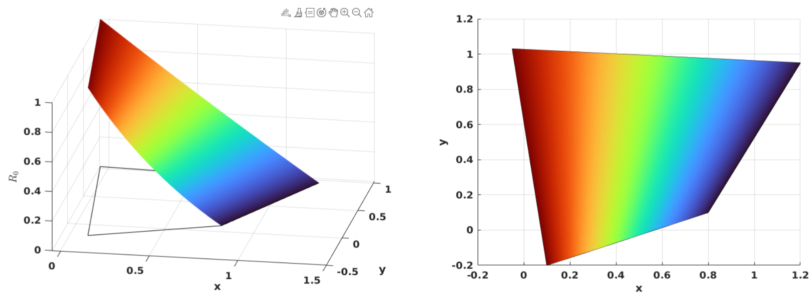

Figure 1.3: Rational function R_0.

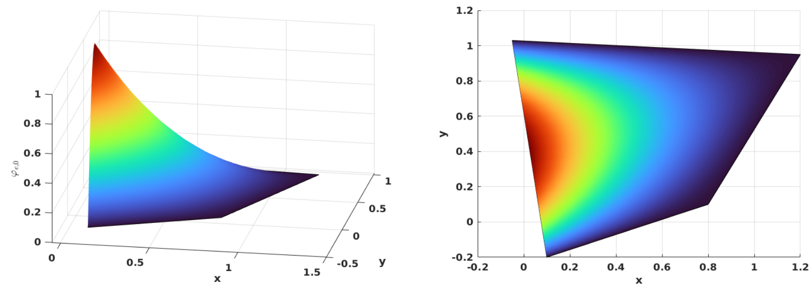

Figure 1.4: Edge nodal basis function $\varphi_{e,0}$ in $\mathcal{DS}_2$.

where $\mathbf{x}_{v,i}$ is the coordinates of the vertex node v_i. The i index is understood in the sence of module of 4 as well. In Figure 1.5, we show $\varphi_{v,0}$ on a general quadrilateral.

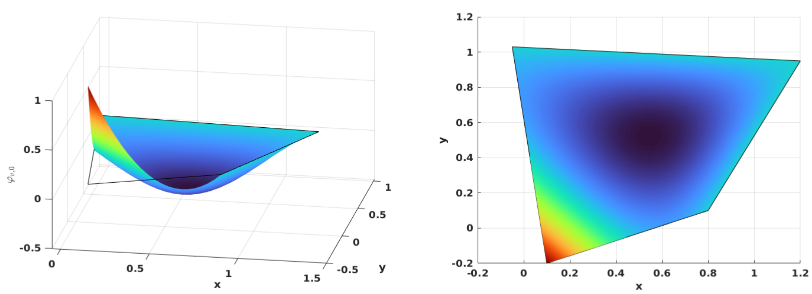

Figure 1.5: Vertex nodal basis function $\varphi_{v,0}$ in $\mathcal{DS}_2$.

We conclude that unisolvence follows directly from the constructed nodal basis functions.

1.2.2 Direct Serendipity Space $\mathcal{DS}_1$

The supplemental space for $\mathcal{DS}_1$ differs from that for $\mathcal{DS}_2$. We present the first order serendipity as

$$\mathcal{DS}_1(E) = \mathbb{P}_1(E) \oplus \operatorname{span}\{R\}, \tag{1.13}$$

where $R(\mathbf{x}_{v,0}) = R(\mathbf{x}_{v,2}) = -R(\mathbf{x}_{v,1}) = -R(\mathbf{x}_{v,3}) = 1$. The construction of rational funcion R is not unique. We present one way to construct $R(\mathbf{x})$ with edge nodal basis function (1.11) in $\mathcal{DS}_2$ as

$$R(\mathbf{x}) = r(\mathbf{x}) - \sum_{i=0}^{3} r(\mathbf{x}_{e,i})\varphi_{e,i}(\mathbf{x}), \tag{1.14}$$

where auxiliary function $r(\mathbf{x})$ is defined as

$$r(\mathbf{x}) = \frac{\lambda_2(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_2(\mathbf{x}_{v,0})\lambda_3(\mathbf{x}_{v,0})} - \frac{\lambda_0(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,1})\lambda_3(\mathbf{x}_{v,1})} + \frac{\lambda_0(\mathbf{x})\lambda_1(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,2})\lambda_1(\mathbf{x}_{v,2})} - \frac{\lambda_1(\mathbf{x})\lambda_2(\mathbf{x})}{\lambda_1(\mathbf{x}_{v,3})\lambda_2(\mathbf{x}_{v,3})}. \tag{1.15}$$

In Figure 1.6, we plot the rational function on a general quadrilateral.

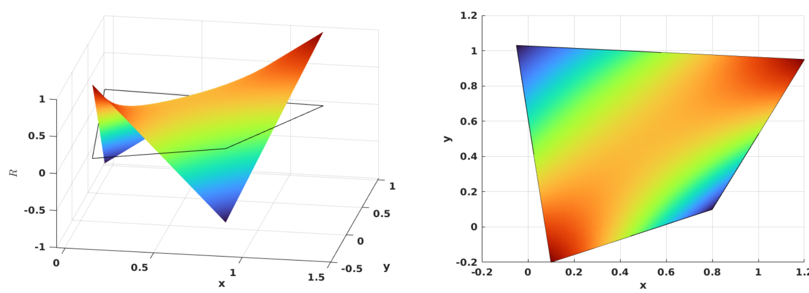

Figure 1.6: Rational function in $\mathcal{DS}_1$.

We define vertex nodal basis function accordingly as

$$\psi_{v,i}(\mathbf{x}) = \frac{\lambda_{d,m}(\mathbf{x}) - \frac{1}{2}\lambda_{d,m}(\mathbf{x}_{v,i+2})(1 + (-1)^i R(\mathbf{x}))}{\lambda_{d,m}(\mathbf{x}_{v,i}) - \lambda_{d,m}(\mathbf{x}_{v,i+2})}, \tag{1.16}$$

where $m \equiv i + 1 \pmod 2$, and index i is understood in the sence of modulo of 4. In Figure 1.7, we show the vertex nodal basis function $\varphi_{v,0}$ in $\mathcal{DS}_1$ on a general quadrilateral, which evaluates 1 at vertex v_0 and 0 at other vertex.

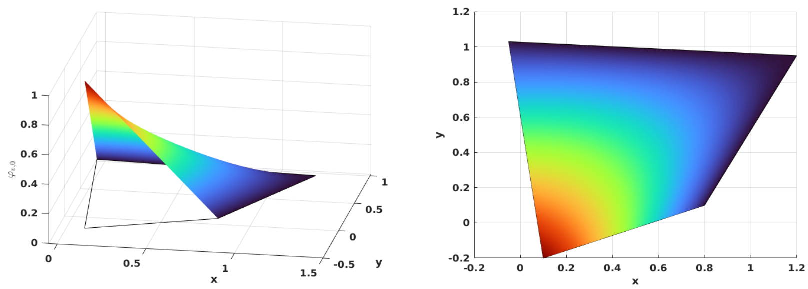

Figure 1.7: Vertex nodal basis function in $\mathcal{DS}_1$.

Lemma 1.1. *Let $\mathcal{T}_h$ be uniformly shape regular* (1.2) *with shape regularity parameter σ_*. For each element $E \in \mathcal{T}_h$, There exists a bounded interpolation operator $\mathcal{I}_h : W_p^l(E) \to \mathcal{DS}_r$. For $r \geq 1$, $E \in \mathcal{T}_h$, $1 \leq q \leq \infty$, and any nonnegative integer m,*

$$\|\mathcal{I}_h^r v\|_{W_q^m} \leq C(\sigma_*, m, q) \sum_{k=0}^{l} h_E^{k-m+\frac{2}{q}-\frac{2}{p}} |v|_{W_p^k(E^*)}, \tag{1.17}$$

where $E^* = \cup_{F \in \mathcal{T}_h, F \cap E \neq \emptyset}$ and $|\cdot|_{W_p^k}$ is the seminorm of k-th order derivatives.

Lemma 1.2. *Under the asumptions of 1.1, there exists a constant $C = C(r, \sigma_*) > 0$ such that for all functions $v \in W_{l,p}(E^*)$, with $1 \leq p \leq \infty$ and $l > 1/p$ or ($l \geq 1$ if $p = 1$),*

$$\|v - \mathcal{I}_h^r\|_{W_{m,p}} \leq C h_E^{l-m} |v|_{W_{l,p}(E^*)}, \quad 0 \leq m \leq \min(l, r+1). \tag{1.18}$$

The proof of lemma 1.1 and lemma 1.2 can be found in Arbogast et al. (2022).

1.3 Mixed Finite Element

We define H^1 and $H(\text{div})$ conforming mixed element space with the direct serendipity space, and test with simple stokes and darcy problems respectively.

1.3.1 $H(\textbf{div})$ Conforming Element

We have some choice for $H(\text{div})$ confirming space. For example, we can choose the famous Ravier-Thomas element by Raviart and Thomas (1977). The serendipity space is the precursor of the Brezzi-Douglas-Marini element by Brezzi et al. (1985). We recover Arbogast-Correa element by Arbogast and Correa (2016) if we use mapped direct serendipity space. We generated a reduced $H(\text{div})$ approximation, that $\nabla \cdot \mathbf{u}$ is approximated one less order than approximation of $\mathbf{u}$.

We write the rotated 2D exact sequence for energy space as

$$H^1 \xrightarrow{\text{curl}} H(\text{div}) \xrightarrow{\text{div}} L^2 \tag{1.19}$$

$$\cup \qquad\qquad \cup \qquad\qquad \cup \tag{1.20}$$

$$\mathcal{DS}_2 \xrightarrow{\text{curl}} \mathbf{V}_r^{r-1} \xrightarrow{\text{div}} \mathbb{P}_{r-1}. \tag{1.21}$$

We define the first order reduced $H(\text{div})$ conforming space with $\mathcal{DS}_2$ as

$$\mathbb{V}_1^0 = \mathbb{P}_1^2 \oplus \text{curl}\mathbb{S}_2^{\mathcal{DS}}. \tag{1.22}$$

The reduced space $\mathbb{V}_1^0$ can be given a global basis with the normal components of the shape functions merged continuously on the shared edge of two elements. First, we write curl of the auxiliary rational function R_i (1.8) as

$$\text{curl}R_i(\mathbf{x}) = \frac{\boldsymbol{\tau}_{i+2}(\mathbf{x})\lambda_i(\mathbf{x}) - \boldsymbol{\tau}_i(\mathbf{x})\lambda_{i+2}(\mathbf{x})}{(\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x}))^2}, \tag{1.23}$$

where the index i is understood in with modulo of 4. On each edge, we define two independent basis functions. $\boldsymbol{\varphi_{l,i}}$ and $\boldsymbol{\varphi_{c,i}}$, such that

$$\boldsymbol{\varphi}_{l,i} \cdot \boldsymbol{\nu}_j|_{e_j} = \begin{cases} 0, & j \neq i, \\ l_i(\mathbf{x}), & j = i, \end{cases} \tag{1.24}$$

$\boldsymbol{\varphi_{l,i}}$ and $\boldsymbol{\varphi_{c,i}}$, such that

$$\boldsymbol{\varphi}_{c,i} \cdot \boldsymbol{\nu}_j|_{e_j} = \begin{cases} 0, & j \neq i, \\ 1, & j = i, \end{cases} \tag{1.25}$$

where l_i is a non vanishing linear polynomial defined on edge e_i orthogonal to one, i.e., $\int_{e_i} l_i ds = 0$. We present a construction of these two basis functions.

$$\tilde{\boldsymbol{\varphi}}_{l,i} = \text{curl}(\lambda_{i+1}\lambda_{i+3}R_i) = (\boldsymbol{\tau}_{i+1}\lambda_{i+3} + \lambda_{i+1}\boldsymbol{\tau}_{i+3})R_i + \lambda_{i+1}\lambda_{i+3}\text{curl}R_i. \tag{1.26}$$

We then normalize it as

$$\boldsymbol{\varphi}_{l,i} = \frac{\tilde{\boldsymbol{\varphi}}_{l,i}(\mathbf{x})|\mathbf{x}_i - \mathbf{x}_{i+3}|}{4\lambda_{i+1}(\mathbf{x}_{\mathbf{e},\mathbf{i}})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{\mathbf{e},\mathbf{i}})}, \tag{1.27}$$

where index is understood in the sense of modulo 4. In Figure 1.8, we draw $\boldsymbol{\varphi}_{l,i}$ on two element sharing a edge. The construction of a the constant basis function is a bit more complicated.

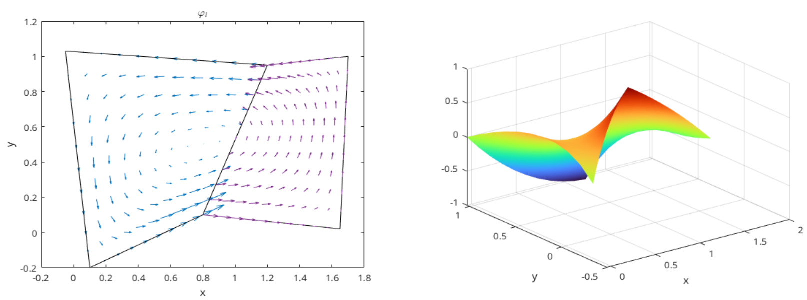

Figure 1.8: Linear basis function $\boldsymbol{\varphi}_l$.

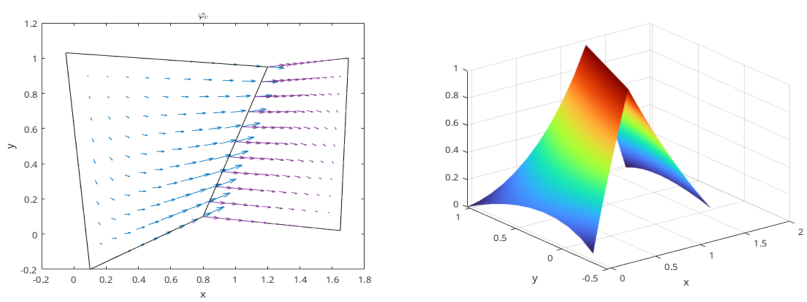

Figure 1.9: Constant basis function $\boldsymbol{\varphi}_c$.

1.3.2 Test Darcy Problem

We test our $H(\mathrm{div})$ conforming element on a simple darcy problem as

$$\begin{aligned} \mathbf{u} + \nabla p &= \mathbf{g} \\ \nabla \cdot \mathbf{u} &= 0 \end{aligned} \tag{1.28}$$

weak form

$$\begin{aligned} (\mathbf{u}, \mathbf{v})_\Omega - (p, \nabla \cdot \mathbf{v})_\Omega &= (\mathbf{g}, \mathbf{v})_\Omega - \langle p_0, v_n \rangle_{\Gamma_i} - \langle \mathbf{u}_0, \mathbf{v} \rangle_{\Gamma_u} \\ (\nabla \cdot \mathbf{u}, \omega) &= 0 \end{aligned} \tag{1.29}$$

Test space $\{\mathbf{v} \in H(\text{div}, \Omega) : v_n = 0 \text{ on } \Gamma_u\}$.

1.3.3 H^1 Conforming Element

Stable mixed finite element pair for Stokes problem has been widely researched. We are referring to the element discussed by Christine Bernardi (1985) and by Fortin (1981). A more general type of element was discussed by Arbogast and Wheeler (2005).

Vertex basis mapping to sobolev space difficulty. Need Clément interpolation. Define Π operator two parts.

We consider the following space for each element $E \in \mathcal{T}_h$ as

$$V_1^{\mathcal{BR}}(E) = (\mathcal{DS}_1(E))^2 \oplus \text{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}, \tag{1.30}$$

where $\mathbf{b}_i$ is a quadratic bubble function defined on each edge e_i. We pick previously defined function $\varphi_{e,i} \in \mathcal{DS}_2$ as

$$\mathbf{b}_i = \varphi_{e,i}(\mathbf{x})\mathbf{n}_i \tag{1.31}$$

which is quadratic when restricted to edge e_i.

Lemma 1.3. *For a element $E \in \mathcal{T}_h$, $\mathbf{u}_h$ is uniquely defined by the following degrees of freedom:*

(i) for corner point v_i and Cartesian direction unitnormal vectors $\{\mathbf{i}, \mathbf{j}\}$,

$$DOF^{(i)}_{(v_i,\mathbf{i})}(\mathbf{u}_h) = \mathbf{u}_h(v_i) \cdot \mathbf{i}, \quad DOF^{(i)}_{(v_i,\mathbf{j})}(\mathbf{u}_h) = \mathbf{u}_h(v_i) \cdot \mathbf{j};$$

(ii) for edge e_i,

$$DOF^{(ii)}_{e_i} = \langle \mathbf{u}_h \cdot \boldsymbol{\nu}_i, 1 \rangle,$$

Proof. We prove the lemma by construction. We pick nodal badis on vertex v_i in two Cartesian direction as

$$N_{v_i,x} = \begin{bmatrix} \psi_{v,i}(\mathbf{x}) \\ 0 \end{bmatrix}, \quad N_{v_i,y} = \begin{bmatrix} 0 \\ \psi_{v,i}(\mathbf{x}) \end{bmatrix} \tag{1.32}$$

We pick edge basis on edge e_i as

$$N_e = \varphi_{e,i}\boldsymbol{\nu}_i \tag{1.33}$$

Note that bubble functions are quadratic when restricted to edge e_i, while nodal basis functions are linear when restricted to edge e_i, therefore bubble functions cannot be in the same space as nodal basis function. We conclude that this element is well-defined, i.e., the degrees of freedom uniquely determine the function. $\square$

We need a π operator, which mappes $(H^1(E))^2$ onto $V_1^{\mathcal{BR}}$ for our spaces. We

1.3.4 Test Stokes Problem

We test our modified BR element on a test Stokes problem. We begin with the strong form of the static Stokes problem as

$$\begin{aligned} -\Delta \mathbf{u} + \nabla p &= \mathbf{f}, \\ \nabla \cdot \mathbf{u} &= 0. \end{aligned} \tag{1.34}$$

Let test function space

$$V = \{\mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \quad \text{on} \quad \Gamma_u\} \tag{1.35}$$

We derive weak form

$$\begin{aligned} (\nabla \mathbf{u}, \nabla \mathbf{v})_\Omega - (p, \nabla \cdot \mathbf{v})_\Omega = (\mathbf{f}, \mathbf{v})_\Omega + \langle \tau^T \nabla \mathbf{u} \nu, \mathbf{v} \cdot \tau \rangle_{\Gamma_i}, \\ + \langle \nu^T \nabla \mathbf{u} \nu - p_0, \mathbf{v} \cdot \nu \rangle_{\Gamma_i} - \langle \nabla \mathbf{u}_0, \nabla \mathbf{v} \rangle_{\Gamma_u} \\ (\nabla \cdot \mathbf{u}, \omega) = 0. \end{aligned} \tag{1.36}$$

Let stress $\mathbf{s} \cdot \tau = \tau^T \nabla \mathbf{u} \nu$ and $\mathbf{s} \cdot \nu = \nu^T \nabla \mathbf{u} \nu - p_0$, we have stress boundary condition term defined on Γ_i which is $< \mathbf{s}, \mathbf{v} >_{\Gamma_i}$. We employ a test space

$$\begin{aligned} (\nabla \mathbf{u}_h, \nabla \mathbf{v}_h)_{\Omega_h} - (p_h, \nabla \cdot \mathbf{v}_h)_{\Omega_h} = (\mathbf{f}, \mathbf{v}_h)_{\Omega_h} + \langle \mathbf{s}, \mathbf{v}_h \rangle_{\Gamma_{i,h}} - \langle \nabla \mathbf{u}_{0,h}, \nabla \mathbf{v}_h \rangle_{\Gamma_{u,h}}, \\ (\nabla \cdot \mathbf{u}, \omega) = 0. \end{aligned} \tag{1.37}$$

We can separate normal and tangent part of stress on the boundary easily when physical boundary align with x-y axis. In the situation when a rectangular computational domain is used, we can mix stress and dirichlet boundary condition even further.

1.3.5 Coupled Degenerate System

We defined scaled formulation (1.38), (1.39), (1.40), and (1.41) in **??**. We follow the procedure by **?** to discretize the strong form.

$$\begin{aligned}
&\check{\mathbf{v}}_r + \phi^{1+\Theta}\nabla(\phi^{-1/2}\check{q}_l) &&= 0, &(1.38)\\
&\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r) + \frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) &&= 0, &(1.39)\\
&\nabla\check{q} - \nabla\cdot\check{\tau}(\check{\mathbf{v}}_s) &&= -(1-\phi)\mathbf{n}, &(1.40)\\
&\nabla\cdot\check{\mathbf{v}}_s - \frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) &&= 0, &(1.41)
\end{aligned}$$

To begin weith, a new Hilbert space is defined for this specific scaled formulation, where the scaled velocity $\check{\mathbf{v}}_r$ should lie in, so

$$\mathbb{V}_d^{\phi} = H_{\phi}(\mathrm{div};\Omega) = \Big\{\mathbf{v}\in(L^2(\Omega))^2 : \phi^{-1/2}\nabla\cdot(\phi^{1+\Phi}\mathbf{v})\in L^2(\Omega)\Big\}, \tag{1.42}$$

which equipped with the inner product as

$$(\mathbf{u}_{\phi},\mathbf{v}_{\phi})_{\mathbb{V}_r^{\phi}} = (\mathbf{u}_{\phi},\mathbf{v}_{\phi}) + (\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{u}_{\phi}), \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v}_{\phi})). \tag{1.43}$$

The normal trace on $\partial\Omega$ is well-defined as well as

$$\left\|\phi^{1/2+\Theta}\mathbf{v}_{\phi}\cdot\nu\right\|_{-1/2,\partial\Omega} < C_{\Omega}\Big\{\,\|\mathbf{v}_{\phi}\| + \left\|\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v}_{\phi})\right\|\,\Big\}. \tag{1.44}$$

Now we define the function spaces

$$\begin{aligned}
\mathbb{V}_d^{\phi} &= \Big\{\mathbf{v}\in H_{\phi}(\mathrm{div};\Omega) : \phi^{1/2+\Theta}\mathbf{v}\cdot\boldsymbol{\nu} = 0 \text{ on } \Gamma_{d,u}\Big\},\\
\mathbb{W}_d &= L^2(\Omega),\\
\mathbb{V}_s &= \Big\{\mathbf{v}\in(H^1(\Omega))^2 : \mathbf{v} \text{ on } \Gamma_{s,u}\Big\},\\
\mathbb{W}_s &= L^2(\Omega),
\end{aligned}\tag{1.45}$$

where $\Gamma_{d,u}\in\partial\Omega$ and $\Gamma_{s,u}\in\partial\Omega$ are boundary assigned with essential boundary condition with nonzero measure.

Scaled nondimensionalized weak form. Find $\check{\mathbf{v}}_r \in \mathbb{V}_r^{\phi} + \check{\mathbf{g}}_r$, $\check{q}_l \in \mathbb{W}_d$, $\check{\mathbf{v}}_s \in \mathbb{V}_s + \mathbf{g}_s$, and $q \in \mathbb{W}_s$ such that

$$\begin{aligned}(\check{\boldsymbol{\tau}}(\mathbf{v}_s), \nabla \boldsymbol{\psi}_s) - (\check{q}, \nabla \cdot \boldsymbol{\psi}_s) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_s),\\ (\nabla \cdot \check{\mathbf{v}}_s, \omega) + \left(\frac{\hat{\phi}}{1-\phi}\check{q}, \omega\right) - \left(\frac{\hat{\phi}^{1/2}}{1-\phi}\check{q}_l, \omega\right) = 0,\\ (\check{\mathbf{v}}_r, \boldsymbol{\psi}_r) - (\check{q}_f, \hat{\phi}^{-1/2}\nabla \cdot (\phi_f^{1+\Theta}\check{\mathbf{v}}_r)) = 0,\\ (\hat{\phi}_f^{-1/2}\nabla \cdot (\phi_f^{1+\Theta}\check{\mathbf{v}}_r), \omega_f) + \left(\frac{1}{\phi_s}\check{q}_f, \omega_f\right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s}\check{q}_h, \omega_f\right) = 0.\end{aligned} \tag{1.46}$$

The Existence and uniquense of the solution has been proved by **?**. We will not repeat the proof here.

1.3.6 MFEM discretization

We discretize the formed degenerate system with previously defined $H(\mathrm{div})$ and H^1 conforming space. We write the function spaces out as

$$\begin{aligned}\mathbb{V}_1^0 &= (\mathbb{P}_1)^2 \oplus \mathrm{curl}\mathbb{S}_2^{\mathcal{DS}}\\ \mathbb{W} &= \mathbb{P}_0\\ \mathbb{V}_1^{\mathcal{BR}} &= \mathcal{DS}_1 \oplus \mathrm{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}\\ \mathbb{W} &= \mathbb{P}_0.\end{aligned} \tag{1.47}$$

We can impose essential boundary conditions

Scaled mixed finite element method. Find $\check{\mathbf{v}}_{r,h} \in \mathbb{V}_1^0 + \hat{\mathbf{g}}_r$, $\check{q})_{r,h} \in \mathbb{W}_D$, $\mathbf{v}_{s,h} \in \mathbf{V}_S + \hat{\mathbf{g}}_s$ and $\check{q}_h \in \mathbb{W}_S$ such that

$$\begin{aligned}(2\phi_s\check{\sigma}(\mathbf{v}_{s,h}), \nabla\Psi_s) - (\check{q}_h, \nabla \cdot \Psi_s) = -(\phi_s\check{\mathbf{f}}, \Psi_s),\\ (\nabla \cdot \check{\mathbf{v}}_{s,h}, \omega) + \left(\frac{\hat{\phi}_f}{\phi_s}\check{q}_h, \omega\right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s}\check{q}_{f,h}, \omega\right) = 0,\\ (\check{\mathbf{v}}_{r,h}, \Psi_r) - (\check{q}_{f,h}, \hat{\phi}_f^{-1/2}\nabla \cdot (\phi_f^{1+\Theta}\check{\mathbf{v}}_r)) = 0,\\ (\hat{\phi}_f^{-1/2}\nabla \cdot (\phi_f^{1+\Theta}\check{\mathbf{v}}_r), \omega_f) + \left(\frac{1}{\phi_s}\check{q}_{f,h}, \omega_f\right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s}\check{q}_h, \omega_f\right) = 0.\end{aligned} \tag{1.48}$$

1.3.7 Local Mass Conservation

Before we continue the discussion, we need to make sure the formulation is well-defined by not dividing by zero. The term $\phi^{-1/2}$ in equation causes trouble when there is no melting, i.e., $\phi = 0$. In order to deal with it, we first define a new element-averaged value as

$$\phi_E = \frac{1}{|E|} \int_E \phi(\mathbf{x}) d\mathbf{x}. \tag{1.49}$$

We replace the

1.3.8 Test Run

1.4 Linear System

1.4.1 Saddle Point Problem

1.4.2 Degenerate Coupled Saddle Point Problem

The linear system corresponding to (insert eqcite) has the form

$$\begin{bmatrix} \mathbf{A}_s & -\mathbf{B}_s & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_s^T & \mathbf{C}_s & \mathbf{0} & \mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_d & -\mathbf{B}_d \\ \mathbf{0} & \mathbf{K} & \mathbf{B}_d^T & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{y}_s \\ \mathbf{x}_d \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \\ \mathbf{f}_d \\ \mathbf{g}_d \end{bmatrix}, \tag{1.50}$$

wherein the solution represents the degrees of freem of primary variables with respect to the bases of the finite element spaces. (Insert details of calculation of each blocked matrix). We form a double saddle point system by symmetrically permutate original linear system,

$$\begin{bmatrix} \mathbf{A}_s & \mathbf{0} & -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d & \mathbf{0} & -\mathbf{B}_d \\ \mathbf{B}_s^T & \mathbf{0} & \mathbf{C}_s & \mathbf{K} \\ \mathbf{0} & \mathbf{B}_d^T & \mathbf{K} & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{x}_d \\ \mathbf{y}_s \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \\ \mathbf{g}_s \\ \mathbf{g}_d \end{bmatrix}. \tag{1.51}$$

We group these block matrix to form a concise version,

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}_d \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} -\mathbf{C}_s & -\mathbf{K} \\ -\mathbf{K} & -\mathbf{C}_d \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} -\mathbf{g}_s \\ -\mathbf{g}_d \end{bmatrix}, \tag{1.52}$$

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix}, \tag{1.53}$$

where $\mathbf{A}$ is a positive definite matrix. However, $\mathbf{B}$ is usually a singular matrix when no melting region presents. We will present a modified Uzawa iteration algorithm to deal with this coupled degenerate singular system. Inexact Uzawa algorithm has been introduced in the previous literature Elman and Golub (1994) and Bramble et al. (1997),

Algorithm 1 Preconditioned Inexact Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$ and $\mathbf{y}_0 = \mathbf{0}$
2: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$
3: Obtain $\mathbf{r} = \mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G}$
4: Obtain $\tilde{\mathbf{y}} = \text{argmin}_{\mathbf{x}\in V}||(\mathbf{B}^T\mathbf{A}^{-1}\mathbf{B} - \mathbf{C})^{-1}\tilde{\mathbf{y}} - \mathbf{r}||$
5: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$

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